

MongoDB Host Access from Jupyter

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Some students have had trouble connecting to their MongoDB instance from a notebook after moving to the current platform. The most common reason is that the new platform uses a different network configuration than the previous one.

! Short Version

Do not connect to `localhost`. Use the MongoDB host address provided by the platform.

Find the MongoDB Host

In the platform, click the **Jupyter** button.

The screenshot shows the JupyterLab interface. At the top, there is a dark blue header bar. Below it, a navigation bar contains the 'Jupyter' button, which is circled in red, and the 'Project' button. To the right of the navigation bar is the 'Jupyter Dash' button. Below the navigation bar is a menu bar with options: File, Edit, View, Run, Kernel, Tabs, Settings, and Help. The main area is divided into three panes. The left pane shows a file browser with a list of files and folders. The middle pane shows a file explorer with a table of files and folders. The right pane shows a notebook titled '071-meet-ds-lab-applicants.ipynb'. The notebook content includes a section titled '1. Meet the DS Lab A' and a code cell with the text '# test'.

Name	Modified
images	last mo.
071-meet-ds-lab-applicants.ip...	yesterday
072-etl-class.ipynb	11 days ago
073-chi-square-test.ipynb	11 days ago
074-dashboard.ipynb	11 days ago
075-assignment.ipynb	11 days ago
business.py	28 days ago
database.py	28 days ago
display.py	28 days ago

1. Meet the DS Lab A

When you decided to start down the path to becoming a data scientist, the first thing you did was to register an account with the Data Science Lab, and began your data science journey! The account takes the admissions exam. Is there a way to predict if an applicant will be accepted into the Data Science Lab?

In this project, you'll help run an experiment to see if the type of experiment can increase the likelihood that they'll be accepted. This type of experiment is called a **hypothesis test** or **statistical test**.

In this lesson, we'll try to get a better sense of what the Data Science Lab — where they're from, how old they are, and more.

Data Ethics: This project is based on a real experiment conducted in June of 2022. There is, however, a concern that the data science team used real student data, which could expose any actual personal data — like name, email address, or phone number — that would violate our students' privacy.

```
[1]: # test
```

This opens a pop-up with the Jupyter and MongoDB connection details. Copy the MongoDB IP address shown there.

In your notebook, store that address in a variable:

```
host = "your-mongodb-ip-address"
```

Then use the `host` variable when you create the MongoDB client:

```
from pymongo import MongoClient

client = MongoClient(host=host, port=27017)
```

Using a variable makes the notebook easier to update. If the IP address changes later, you only need to update the `host` value in one place.

Common Error: Connecting to Localhost

If you see an error like this, your code is trying to connect to `localhost`:

```
ServerSelectionTimeoutError: localhost:27017: [Errno 111] Connection refused
```

The important part is `localhost:27017`. That means the notebook is looking for MongoDB on the notebook machine itself, not on the MongoDB host provided by the platform.

This usually happens when the code still looks like this:

```
client = MongoClient(host="localhost", port=27017)
```

Change it to:

```
client = MongoClient(host=host, port=27017)
```

Why `host=host` Is Correct

The expression `host=host` can look confusing at first because the same word appears twice.

In this line:

```
client = MongoClient(host=host, port=27017)
```

The first `host` is the parameter name expected by `MongoClient`.

The second `host` is the variable you created earlier:

```
host = "your-mongodb-ip-address"
```

So `host=host` means: pass the value stored in the `host` variable to the `host` parameter.

Another Common Mistake: Using "host" as Text

If your error message mentions something like `host:27017`, check whether your code uses quotation marks around `host`:

```
client = MongoClient(host="host", port=27017)
```

That is different from `host=host`.

"host" is just the literal text `host`. Python will not replace it with your variable value.

Use this instead:

```
client = MongoClient(host=host, port=27017)
```

How to Read the Error Message

Error messages usually include useful clues. For MongoDB connection issues, look for the host and port in the message:

- `localhost:27017` usually means the code is still using `localhost`.
- `host:27017` usually means the code is using the string "host" instead of the variable `host`.
- A platform IP followed by `:27017` means the code is at least trying the intended host, so the next thing to check is whether the copied IP address is correct.

The error type is also useful. `ServerSelectionTimeoutError` means the client could not reach a MongoDB server before the timeout. In this case, the cause is often the wrong host address.

When You Still Need Help

If you have checked the host value and still cannot connect, reach out to the support team or attend an Office Hours session. Include the exact error message and the connection code you are using, but do not share private credentials.